

# ZAREEN ZIA

Senior Software Engineer |  
AI & Data Engineering |  
MSc in Data Science



01791396479



Uttara, Dhaka, Bangladesh



zareenzia801@gmail.com



<https://github.com/zareenzia>



<https://www.linkedin.com/in/zareenzia801/>

## TECHNICAL SKILLS

### Languages

- Java
- Python
- SQL

### Backend

- Spring Boot
- Spring Security
- Spring Data JPA
- Hibernate
- REST APIs
- Microservices

### Data & AI

- Machine Learning
- NLP
- RAG
- Regression
- Prompt Engineering
- Classification
- Clustering
- Pandas
- Data Analytics
- Data Mining

### Database

- PostgreSQL
- MySQL
- Google Cloud SQL
- BigQuery

### AI Tools

- ChatGPT
- Claude
- Gemini
- GitHub Copilot
- OpenAI API (learning)
- AI Agents (learning)

### Tools

- Git
- IntelliJ IDEA
- Maven & Gradle
- JUnit 5
- Linux,
- Jira,
- Confluence,
- Postman

## ABOUT ME

Senior Software Engineer with 4+ years of experience developing enterprise backend applications using Java, Spring Boot, REST APIs, and cloud technologies. Pursuing an MSc in Data Science with expertise in Machine Learning, Data Analytics, and SQL. Passionate about AI, LLM-powered applications, and building scalable data-driven solutions through modern software engineering practices.

## PROFESSIONAL EXPERIENCE

### Senior Software Engineer BJIT Limited

April, 2022 - Present

Promoted from Software Engineer (June 2025)

- Developed scalable backend applications using Java, Spring Boot, and Microservices.
- Built 50+ secure RESTful APIs with cloud integrations (GCP, AWS) and Docker deployments.
- Improved application performance and maintained 90%+ unit test coverage.
- Owned feature development, performed code reviews, and mentored junior developers.
- Collaborated with 30+ engineers in Agile Scrum teams to deliver enterprise solutions.
- **Key projects:** PT Console Ventana (GCP, BigQuery, Cloud SQL, AWS SES), Rakuten ECSG LABO & Rakuten FMCG (Spring MVC/BFF architecture), and Denka digital transformation.

## EDUCATION

### Master of Science (MSc) in Data Science United International University , Dhaka

Expected Graduation: 2027

Current CGPA: 3.70 / 4.00

### BSc. in Computer Science and Engineering

American International University of Bangladesh, Dhaka

Passing Year: 2021

CGPA: 3.58 / 4.00

- Intern as a Teaching Assistant in Web Technology course.

## PROJECTS

### Financial AI Application (In Progress)

**Technologies:** Java, Spring Boot, PostgreSQL on NeonDb, React, REST API, Railway

Description: Developing an AI-powered personal finance management application

Features:

- Transaction Management
- Budget Tracking
- Analytics Dashboard
- Notes
- Categories
- Theme Customization
- AI-powered Financial Insights (Planned)
- LLM Integration (Planned)

GitHub: <https://github.com/zareenzia/FinTrack>

## RESEARCH PROJECTS

### PulmoRAG — Hybrid RAG System for Pulmonary Disease Information

**Technologies:** Python, ChromaDB, BM25, Nomic Embed, Cross-Encoder Reranking,

LLaMA-3.1/GPT-4/Claude, Streamlit, Docker

- Built a hybrid retrieval-augmented generation pipeline (dense + sparse retrieval with Reciprocal Rank Fusion and cross-encoder reranking) indexing 7,409 chunks from 22 clinical guidelines (~60MB).
- Achieved a 0.858 composite RAGAS score across 102 test queries; implemented relevance detection for out-of-scope queries at 92% accuracy.
- Supported both local (LLaMA) and cloud (OpenAI, Anthropic) LLM inference; fully containerized with Docker.

### Multi-Layered Context-Aware Smart City Navigation Framework for the Visually Impaired

**YOLOv8, Raspberry Pi, IoT (GPS-GSM, BLE), NLP-TTS**

- Designed a multimodal IoT assistive-navigation system combining real-time object detection, ultrasonic sensing, and audio/haptic feedback.
- Achieved 74.1% mAP@0.5 object detection accuracy across indoor/outdoor smart-city scenarios.

### Restoration of Old and Faded Documents Using Digital Image Processing

**Python, OpenCV, Tesseract OCR**

- Built an explainable image-restoration pipeline (CLAHE, adaptive thresholding, morphology) for degraded and faded documents.
- Improved OCR confidence from 40.3 to 51.5 and contrast by 30.8% on faded receipt images.

## CERTIFICATIONS

Information Security Training (ISMS) by BJIT Academy Limited.

IT Essentials course - Cisco Networking Academy

Technical Skill Training (held between 1st April to 25th July 2022 ) by BJIT Academy

Spring Framework Master Class – Java Spring the Modern Way by Udemy.